

Crash Course in Fast Food

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Behind the Wheel, beyond the Meal: Anecdotal evidence or a real danger?



From Bites to Bytes: Building Our Dataset

Source for Fast-Food

- Open Street Map & Overpass API
- Extraction of (fast-food) coordinates and road features (e.g. traffic lights, stop signs)
- Format: name/item & coordinates for different subcategories of places
 - Food-through-window Drive-thrus
 - Curbside pickup

Source for Crashes

- Public records request to Ohio Department of Public Safety
 - csv file: 1.289.913 crashes in Ohio between Sep 2019 and Sep 2025 (in Franklin County: 115.682)
 - Most relevant columns: coordinates and "Narrative" (written police report) → is a crash associated with food?

Model: Geospatial Autocorrelation

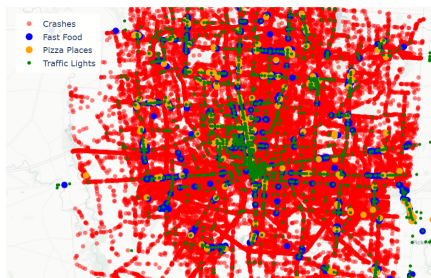


Figure: Franklin county map

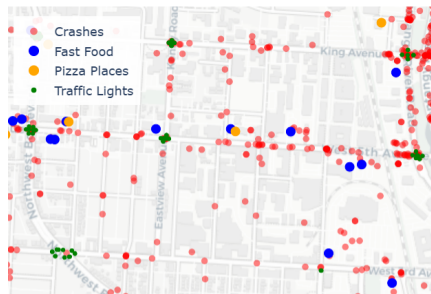


Figure: Zoomed in to 5th Avenue

Geographically Weighted Regression (GWR)

$$D_{\text{Crash}}(x, y) = a(x, y) + b(x, y) \cdot D_{\text{Fastfood}}(x, y) + c$$

Beyond the Drive-Thru: What the Data Says

- Moran's I analysis helps detect clusters and outliers

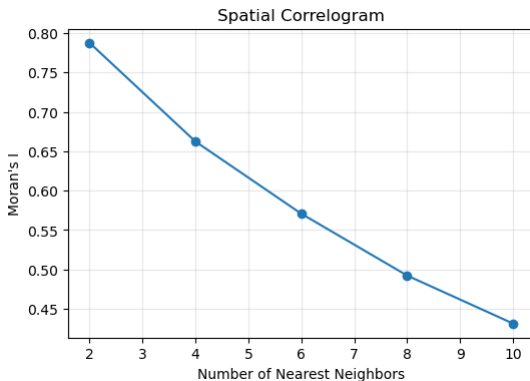


Figure: Spatial correlogram using Moran's I as a measure of the likelihood of clustering of certain events for the fast food dataset.

Correlating crash sites to fast food locations

Local GWR Coefficient: Fastfood Count vs. Crash Density

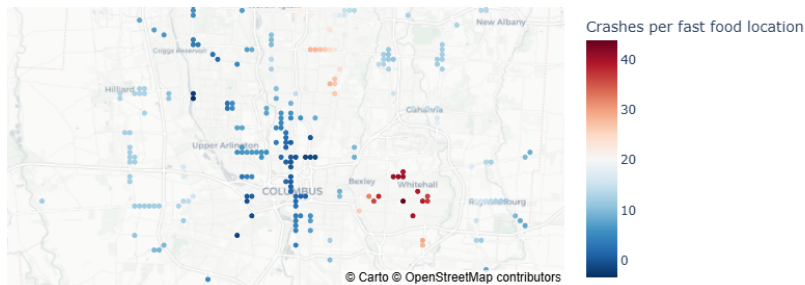


Figure: Grid of crash locations and fast food chain proximity correlations.

Correlating crash sites to pizza restaurant locations

Local GWR Coefficient: Pizza Restaurant Count vs. Crash Density

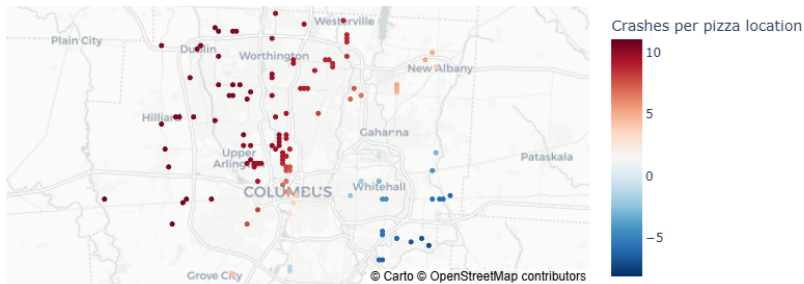


Figure: Grid of crash locations and pizza restaurant chain proximity correlations.

- Further considerations: Top 10 coffee chains
 - Does the need for caffeine play a role?
- Further considerations: Top 10 gas stations
 - Similar road situation, but less distracted because of food

Key Takeaways

- Car crash occurrences seem to be correlated to fast food restaurant proximity
- But: Restaurants often serve as landmarks in the crash reports
- OpenStreetMap data is vulnerable to inaccuracies



Figure: 'Drive-thru' taken too literally :-)